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Optimizing Discount and Marketing Strategies to Maximize Sales Performance and Profitability

Project Domain

[Excel and Power BI]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT:

This project was carried out as part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, an initiative by Edunet Foundation in collaboration with Microsoft, LinkedIn, and SAP. The program focuses on developing practical skills in Artificial Intelligence, Data Analytics, and Business Intelligence.

The project aims to analyse discount and marketing strategies to improve sales performance and profitability. In today's competitive business environment, companies use discounts and promotional campaigns to attract customers; however, ineffective strategies can negatively impact profit margins. This study focuses on understanding the relationship between discounts, marketing efforts, and business performance.

A structured dataset containing sales-related information such as revenue, cost, profit, region, product, and customer type was used for analysis. Data cleaning and pre-processing techniques were applied to ensure accuracy by removing duplicates and handling missing values. The analysis was performed using Microsoft Excel and Power BI, where tools like Pivot Tables, formulas, and dashboards were used to calculate key performance indicators such as total sales, net profit, and profit margin.

The results of the analysis helped identify high-performing products, regions, and effective marketing strategies. It also provided insights into how discount levels influence customer purchasing behaviour and overall profitability. Based on these findings, the project suggests data-driven strategies to optimize discount usage and improve marketing effectiveness.

Overall, this project highlights how data analytics and business intelligence tools can be used to optimize discount and marketing strategies, leading to improved sales performance and profitability.

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CHAPTER 1

Introduction

This project is carried out as part of the AICW Fellowship Program, which focuses on developing practical skills in Artificial Intelligence, Data Analytics, and Business Intelligence. The project analyzes discount and marketing strategies and their impact on sales performance and profitability. Businesses use discounts and campaigns to increase sales, but improper strategies can reduce profits. Using tools like Microsoft Excel and Power BI, this project examines sales data to identify trends and generate useful insights. This chapter presents the problem statement, motivation, objectives, and scope of the project.

1.1 Problem Statement:

In modern business environments, organizations frequently rely on discounts and marketing campaigns to boost sales and remain competitive. However, these strategies are often implemented without proper data-driven evaluation, leading to situations where increased sales do not necessarily translate into higher profitability. In many cases, excessive discounting reduces profit margins, while ineffective marketing efforts result in wasted resources.

The core challenge lies in understanding the true impact of discounts and marketing strategies on overall business performance. Businesses often lack clear insights into which combinations of discount levels, products, regions, and customer segments generate sustainable profits. This creates a gap between sales growth and financial performance.

This problem affects decision-makers such as sales managers and marketing teams who depend on accurate data insights for strategic planning. Traditional methods often focus only on sales volume rather than profitability, and lack proper analytical tools to evaluate performance comprehensively.

Therefore, there is a need for a data-driven approach that integrates sales analysis, profitability evaluation, and visualization techniques to identify optimal discount and marketing strategies that maximize both sales performance and profit.

1.2 Motivation:

The motivation behind this project is to understand how businesses can improve their sales performance and profitability using data-driven strategies. In many real-world situations, companies focus more on increasing sales through discounts and promotions without analyzing their impact on profit. This creates a need to study how different strategies affect overall business performance.

This project also helps in gaining practical knowledge of data analytics tools such as Microsoft Excel and Power BI, which are widely used in industries. It provides an opportunity to analyse real-world data, identify patterns, and generate meaningful insights.

Additionally, this project aligns with current industry trends where organizations rely on data analytics and business intelligence for decision-making. The insights generated from this analysis can help businesses make better strategic decisions and improve their performance.

1.3 Objectives:

The following objectives are defined for this project:

1. To analyse the impact of discount strategies on sales performance and profitability.
2. To evaluate the effectiveness of marketing campaigns across different regions and products.
3. To identify high-performing products, regions, and customer segments.
4. To clean and pre-process the dataset for accurate and reliable analysis.
5. To create interactive dashboards using Excel or Power BI for better visualization of insights.
6. To provide data-driven recommendations to improve business performance and decision-making.

1.4 Scope of the Project:

- This project focuses on analysing a structured sales dataset containing details such as revenue, cost, profit, product, region, and customer type.
- Microsoft Excel is used for data cleaning, pre-processing, and basic analysis.
- Power BI is used for creating interactive dashboards and visualizations to present insights clearly.
- The project analyses the impact of discount and marketing strategies on sales performance and profitability.
- The analysis includes identifying trends, high-performing products, and regional performance.
- The scope is limited to the available dataset and does not include real-time data analysis.
- Advanced techniques such as predictive modelling or machine learning are not included.
- The project is developed for academic and learning purposes as part of the fellowship program.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research:

Previous studies show that businesses use data analytics tools to improve sales performance and decision-making. Many organizations rely on Microsoft Excel for data cleaning, analysis, and basic reporting. Techniques such as Pivot Tables, filtering, and data validation are commonly used to identify trends and patterns in sales data.

In recent years, tools like Power BI have been widely adopted for advanced data visualization and business intelligence. These tools help organizations create interactive dashboards to monitor key performance indicators such as sales, profit, and customer behavior.

However, many existing approaches focus mainly on increasing sales without considering profitability. Excessive discounts and poorly planned marketing strategies often lead to reduced profit margins. Additionally, lack of proper data visualization and analysis makes it difficult for decision-makers to gain clear insights.

This project aims to address these gaps by analyzing both sales performance and profitability using data-driven techniques and visualization**.**

2.2 Tools and Technologies Review:

The tools and technologies used in project are;

- Microsoft Excel — widely used for data cleaning, transformation, and basic visualization. Pivot tables, conditional formatting, and data validation are commonly used techniques.
- Microsoft Power BI — a business analytics tool that provides interactive visualizations and business intelligence capabilities with an interface simple enough for end users to create their own reports and dashboards.
- Microsoft Copilot (AI) — an AI assistant integrated into Microsoft 365 tools that helps automate tasks, generate summaries, and assist with data analysis using natural language prompts.
- AI Chatbots (e.g., ChatGPT, Gemini) — generative AI tools used for content creation, marketing copy, report generation, and prompt-driven analysis.

2.3 Gaps Identified:

- Most existing studies focus mainly on increasing sales without giving equal importance to profitability, especially in the context of discount strategies.
- There is limited analysis on how different discount levels affect profit margins across products, regions, and customer segments.

- Many approaches rely on basic data analysis and do not use effective visualization tools like Power BI to present insights clearly.
- There is a lack of integrated use of Excel and Power BI for comprehensive analysis and decision-making in student-level projects.
- Existing methods often do not provide clear, data-driven recommendations for optimizing both sales performance and profitability.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design:

The system is designed to analyze sales data and evaluate the impact of discount and marketing strategies on business performance. The process begins with collecting a structured dataset containing details such as revenue, cost, profit, product, region, and customer type.

The data is then cleaned using Microsoft Excel by removing duplicates, handling missing values, and correcting data formats to ensure accuracy. After preprocessing, the data is analyzed using Excel functions like Pivot Tables and formulas to calculate key performance indicators such as total sales, profit, and profit margin.

Further analysis and visualization are carried out using Power BI, where interactive dashboards are created to represent trends, comparisons, and performance metrics. These dashboards help in understanding the relationship between discounts, marketing strategies, and profitability.

Finally, insights are generated to support better decision-making and improve overall business performance.

3.2 Modules / Components Used:

Module 1 – Data Collection:

Loading the dataset into Microsoft Excel and Power BI for analysis.

Module 2 – Data Cleaning:

Removing duplicate records, handling missing values, and standardizing data formats.

Module 3 – Data Analysis:

Applying calculations using Pivot Tables, formulas, and basic analysis techniques to identify trends and patterns.

Module 4 – Data Visualization:

Creating interactive dashboards and charts in Power BI to present insights clearly.

Module 5 – Insight Generation:

Interpreting the results and providing data-driven recommendations for improving sales and profitability.

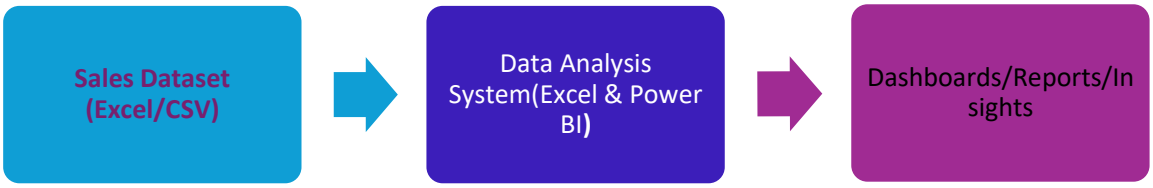
3.3 Data Flow Diagram (DFD):

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram

The Data Flow Diagram (DFD) Level 0 represents the overall system as a single process. It shows how the input data is processed and converted into meaningful outputs.

In this project, the system takes sales dataset as input, processes the data using Microsoft Excel and Power BI, and generates insights, reports, and dashboards as output. This helps in analysing discount and marketing strategies to improve sales performance and profitability.



DFD Level 1 — Detailed Process Flow

The Level 1 Data Flow Diagram provides a detailed view of the system by breaking down the main process into smaller sub-processes. It shows how data moves through different stages such as data loading, cleaning, analysis, and visualization.

In this project, the process starts with loading the sales dataset into Excel and Power BI. The data is then cleaned and transformed by removing duplicates and handling missing values. After that, the data is analyzed using various techniques to identify trends and patterns. Finally, the results are visualized through dashboards and reports to generate meaningful insights.

3.4 Advantages:

The following are the key advantages of the proposed system:

- Provides clear insights into the impact of discount and marketing strategies on sales and profitability.
- Helps in better decision-making through data-driven analysis.
- Reduces manual effort by using Excel for data cleaning and preprocessing.
- Uses Power BI to create interactive dashboards for easy understanding of data.
- Identifies high-performing products, regions, and customer segments.
- Improves visualization of trends and patterns for better business analysis.
- Cost-effective solution using widely available tools like Excel and Power BI.

- Can be easily applied to similar business datasets for future analysis.

3.5 Requirement Specification:

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for downloading tools and accessing online features

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft Excel (2019 +)
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

The data cleaning process was carried out to improve the quality and accuracy of the dataset before analysis.

Dataset Source:

The dataset used in this project is a sales dataset prepared for analyzing discount and marketing strategies.

Original Dataset Size:

The dataset contained multiple rows and columns with raw sales data.

Issues Found:

- Duplicate records
- Missing or null values
- Inconsistent data formats
- Incorrect data types

Cleaning Steps Applied:

- Removed duplicate entries to avoid redundancy
- Handled missing values appropriately
- Standardized data formats for consistency
- Corrected data types for accurate analysis
- Organized and structured the dataset

Cleaned Dataset Size*:*

After cleaning, the dataset became more consistent, accurate, and ready for analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Date	Region	City	Product_Name	Customer_Typ	Total_Sales	Total_Cost	Net_Profit	Discount_Percenta	Payment_Mode	Marketing_Campaign	Customer_Rating	
2	Jan-2025	South	Mumbai	Laptop	Wholesale	459939	313188	146751	8	Card	New Launch	4.1	
3	Jan-2025	South	Mumbai	T-Shirts	Retail	381690	352863	28827	18	Cash	No Campaign	5	
4	Jan-2025	South	Mumbai	Rice Bags	Corporate	78060	32479	45581	11	Card	New Launch	4	
5	Jan-2025	South	Mumbai	Sofa	Corporate	396778	283152	113626	2	Net Banking	Festival Offer	3.5	
6	Jan-2025	South	Hyderabad	Medicines	Retail	838842	702038	136804	11	Card	Clearance Sale	4.9	
7	Jan-2025	North	Mumbai	Laptop	Wholesale	217487	156232	61255	12	Card	Clearance Sale	3.3	
8	Jan-2025	North	Kolkata	T-Shirts	Corporate	647390	476899	170491	13	UPI	Clearance Sale	4.6	
9	Jan-2025	North	Hyderabad	Rice Bags	Corporate	882233	207757	674476	11	Cash	Clearance Sale	4.6	
10	Jan-2025	North	Kolkata	Sofa	Corporate	585787	371456	214331	19	Card	Festival Offer	3.9	
11	Jan-2025	North	Hyderabad	Medicines	Retail	724875	265058	459817	14	Card	Festival Offer	4.5	
12	Jan-2025	East	Kolkata	Laptop	Corporate	511119	276956	234163	5	Net Banking	New Launch	4.4	
13	Jan-2025	East	Mumbai	T-Shirts	Corporate	460331	329548	130783	5	UPI	Festival Offer	4.6	
14	Jan-2025	East	Delhi	Rice Bags	Retail	193392	141703	51689	3	Cash	No Campaign	4	
15	Jan-2025	East	Kolkata	Sofa	Corporate	698135	170268	527867	5	Cash	New Launch	4.1	
16	Jan-2025	East	Hyderabad	Medicines	Corporate	542801	99718	443083	1	Cash	Clearance Sale	3.1	
17	Jan-2025	West	Kolkata	Laptop	Retail	594802	86053	508749	18	Net Banking	Festival Offer	3.9	
18	Jan-2025	West	Kolkata	T-Shirts	Wholesale	132467	83313	49154	19	UPI	No Campaign	3.3	
19	Jan-2025	West	Mumbai	Rice Bags	Wholesale	434463	430761	3702	14	UPI	No Campaign	3.4	
20	Jan-2025	West	Mumbai	Sofa	Retail	186029	134806	51223	1	UPI	No Campaign	4.9	
21	Jan-2025	West	Delhi	Medicines	Corporate	349275	138240	211035	6	UPI	No Campaign	4.4	
22	Feb-2025	South	Delhi	Laptop	Retail	283456	135062	148394	2	Cash	New Launch	3.3	
23	Feb-2025	South	Delhi	T-Shirts	Retail	480095	342430	137665	19	Card	No Campaign	4.3	

Figure 1: Screenshot of the raw (uncleaned) dataset.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Date	Region	City	Product_Name	Customer_Type	Quantity	Unit_Price	Total_Sales	Total_Cost	Net_Profit	Discount_Percentage	Payment_Mode	Marketing_Campaign	Customer_Rating	Validated_Profit
2	2025-08-21	West	Chennai	Medicines	Wholesale	11	17744	195184	153457	41727	10	Cash	No Campaign	4.1	41727
3	2025-12-16	East	Hyderaba	Laptop	Wholesale	2	16288	32576	22269	10307	2	Net Banking	Festival Offer	3.3	10307
4	2025-09-08	West	Hyderaba	Rice Bags	Wholesale	10	30371	303710	236862	66848	21	Card	Summer Sale	3.1	66848
5	2025-11-16	East	Mumbai	Sofa	Wholesale	14	43789	613046	470923	142123	21	Card	Summer Sale	3.6	142123
6	2025-02-05	East	Delhi	Refrigerator	Corporate	1	31551	31551	20167	11384	13	Card	Clearance Sale	4.3	11384
7	2025-12-02	West	Hyderaba	Medicines	Corporate	7	5497	38479	24460	14019	16	Card	No Campaign	3.8	14019
8	2025-08-28	North	Hyderaba	Mobile	Retail	11	49305	542355	357221	185134	6	UPI	Clearance Sale	5	185134
9	2025-01-02	North	Hyderaba	Medicines	Corporate	12	38025	456300	350364	105936	15	Cash	Festival Offer	4.1	105936
10	2025-02-07	North	Chennai	Rice Bags	Corporate	5	32319	161595	102229	59366	5	Net Banking	No Campaign	4.8	59366
11	2025-05-11	West	Bangalore	T-Shirts	Retail	8	38645	309160	213304	95856	15	Net Banking	Festival Offer	4.1	95856
12	2025-01-04	West	Hyderaba	Mobile	Retail	20	1645	32900	21555	11345	23	UPI	New Launch	4.9	11345
13	2025-10-23	East	Delhi	Refrigerator	Retail	4	42465	169860	137377	32483	16	Net Banking	No Campaign	3.9	32483
14	2025-03-10	North	Bangalore	Refrigerator	Corporate	8	49014	392112	253753	138359	23	UPI	No Campaign	4.2	138359
15	2025-03-06	South	Kolkata	T-Shirts	Corporate	11	42418	466598	281870	184728	13	Card	No Campaign	4.2	184728
16	2025-03-17	South	Mumbai	Sofa	Wholesale	12	8021	96252	84126	12126	9	Net Banking	New Launch	4.8	12126
17	2025-11-24	South	Delhi	Laptop	Corporate	5	17695	88475	64770	23705	9	UPI	New Launch	3.3	23705
18	2025-11-07	North	Chennai	Refrigerator	Corporate	7	37636	263452	186162	77290	22	UPI	Clearance Sale	3.9	77290
19	2025-02-13	West	Delhi	Rice Bags	Wholesale	2	7633	15266	13623	1643	22	Cash	Clearance Sale	4.9	1643
20	2025-10-22	South	Bangalore	Mobile	Retail	3	13522	40566	28963	11603	21	Cash	Clearance Sale	4.5	11603
21	2025-09-25	West	Delhi	T-Shirts	Wholesale	14	5671	79394	51852	27542	13	UPI	Clearance Sale	4.1	27542
22	2025-02-26	South	Bangalore	Laptop	Wholesale	10	40794	407940	350406	57534	3	Card	No Campaign	3.5	57534
23	2025-06-14	West	Mumbai	Medicines	Wholesale	5	7723	38615	26014	12601	6	Card	Clearance Sale	3.7	12601
24	2025-05-11	East	Kolkata	Mobile	Corporate	14	48336	676704	490473	186231	15	Card	No Campaign	3	186231
25	2025-07-22	West	Chennai	T-Shirts	Retail	12	19203	230436	169331	61105	10	Cash	Summer Sale	4.6	61105
26	2025-04-08	East	Kolkata	Refrigerator	Corporate	6	33365	200190	136523	63667	1	UPI	New Launch	3.5	63667

Figure 2: Screenshot of the cleaned dataset (After Data Cleaning)

4.2 Visualization / Output Results:

- Dashboard / Chart 1: Displays key KPIs (Total Sales, Total Cost, Total Profit, Average Rating, Profit Margin) using Card visuals for quick performance overview.
- Dashboard / Chart 2: Shows Total Sales by Region using a Bar Chart, comparing sales across regions.
- Dashboard / Chart 3: Represents Profit by Product Name using a Donut Chart, highlighting contribution of each product.
- Dashboard / Chart 4: Illustrates Sales Trend Over Time using a Line Chart, analyzing sales patterns.
- Dashboard / Chart 5: Displays Total Profit by Product Name using a Treemap, identifying top-performing products.

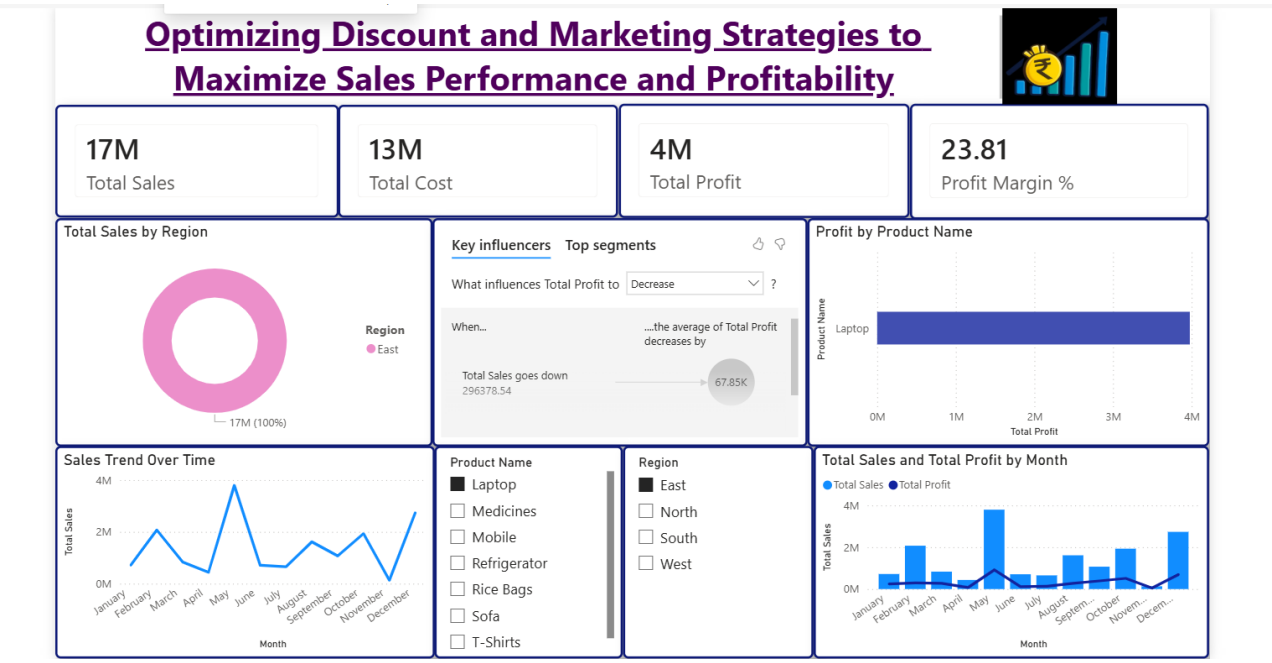


Figure 3: Screenshot of your main dashboard 1 / report / output using POWER BI

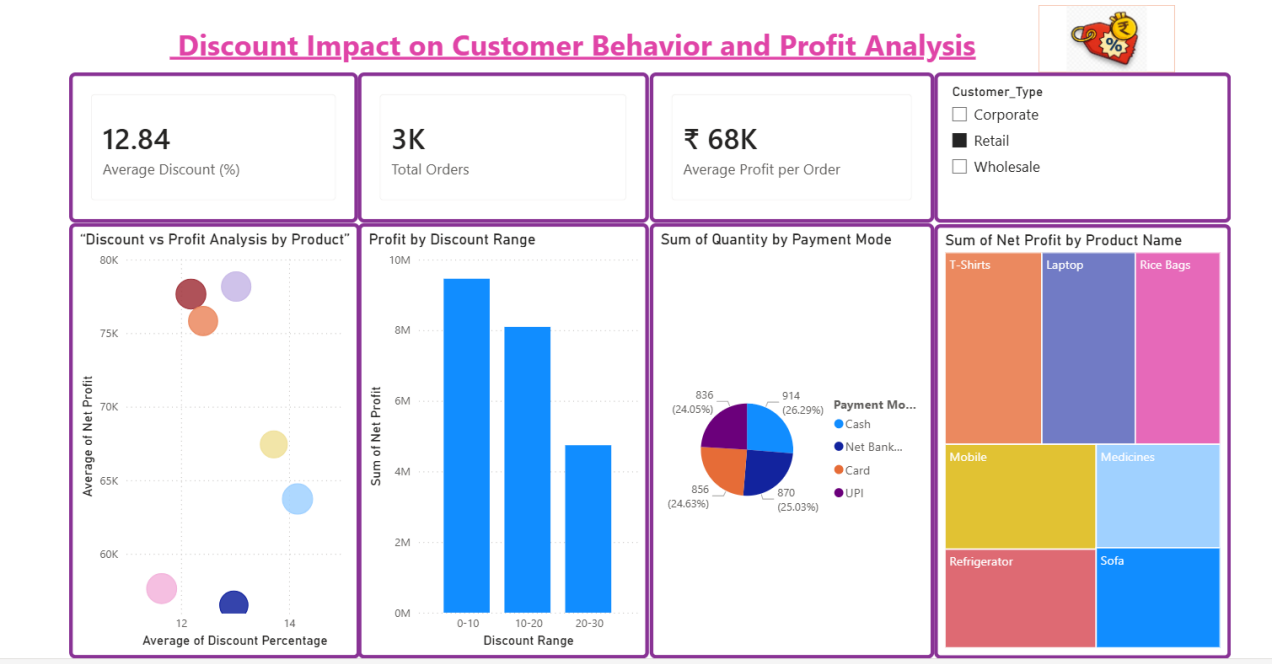


Figure 4: Screenshot of your main dashboard 2 / report / output using POWER BI

4.3 AI Tool Integration Results:

1. **Tool used:** ChatGPT and Microsoft Co-pilot were used to support data analysis, content generation, and dashboard explanation.
2. **Prompts used:**

- “Generate insights from sales and profit data.”
- “Explain Power BI dashboard in simple terms for presentation.”
- “Suggest key findings and conclusions for my project report.”

3. **Output quality:** The AI-generated outputs were clear, structured, and easy to understand. They helped in simplifying complex data insights and improving the overall presentation quality. Minor modifications were made to align outputs with the project context.
4. **How it enhanced the project:** AI tools significantly improved efficiency by reducing the time required for analysis and report writing. They helped generate meaningful insights, improve dashboard explanations, and enhance the overall quality and professionalism of the project.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings:

- After data cleaning, duplicate and inconsistent records were removed, which improved the accuracy and reliability of the analysis.
- Total sales and profit analysis show that a few key products contribute significantly to overall revenue, indicating product concentration.
- Higher discount ranges (20–30%) do not always result in higher profits, suggesting that excessive discounts may reduce profitability.
- Moderate discount ranges (10–20%) generate a better balance between sales volume and profit.
- Sales trends over time indicate fluctuations, with certain months showing peak performance, highlighting seasonal demand patterns.
- Regional analysis reveals that some regions contribute more to total sales than others, helping identify high-performing markets.
- Payment mode analysis shows that digital payments (UPI/Card/Net Banking) are widely used, indicating a shift toward cashless transactions.
- Customer segmentation shows that different customer types (Retail, Corporate, Wholesale) contribute differently to sales and profit.
- The dashboard enables interactive filtering (by product, region, and customer type), making it easier to derive actionable business insights.

5.2 Project Links:

Share the relevant links to your project work for evaluation.

- GitHub Link of the Project:https://github.com/chaphesirisha/Optimizing-Discount-and-Marketing-Strategies-to-Maximize-Sales-Performance-and-Profitability.git
- Video Recording of Project Demonstration:: https://youtu.be/jJxOhHrAaRQ
- Power BI / Dashboard Link : https://app.powerbi.com/links/hhLERj3EWZ?ctid=aed43286-0f66-47d6-942b-d31a4b8addca&pbi_source=linkShare

5.3 Limitations:

The following are the key limitations of the proposed system:

- The project is based on a sample dataset and may not fully represent real-world business scenarios.
- The analysis depends on the quality of the available data; any hidden errors may affect results.
- Real-time data integration is not included in this project.
- The dashboard focuses on historical data and does not include predictive analysis.

- The use of basic tools limits advanced analytical capabilities.

5.4 Future Work:

- Real-time data integration can be implemented to enable continuous monitoring and analysis of sales data.
- Advanced techniques such as forecasting and predictive analytics can be added to improve decision-making.
- The dashboard can be enhanced with more interactive and user-friendly features.
- Additional datasets can be incorporated to provide deeper and more comprehensive insights.
- AI-based tools can be further integrated to automate analysis and generate smarter insights.

5.5 Conclusion:

This project focused on analysing sales data to understand the impact of discount and marketing strategies on business performance. The main objective was to convert raw, unstructured data into meaningful insights using tools like Microsoft Excel and Power BI.

During the project, the dataset was carefully cleaned by removing duplicates, handling missing values, and correcting data formats. After pre-processing, detailed analysis was performed using Excel techniques and Power BI dashboards to identify trends, patterns, and key performance indicators. The final dashboards clearly presented insights related to sales performance, profit trends, and customer behaviour.

The results showed that proper data cleaning and visualization significantly improve decision-making accuracy. The project also demonstrated how business intelligence tools can simplify complex data and make it easy to understand for users. All the defined objectives of the project were successfully achieved.
This project adds value by providing a practical approach to sales data analysis, which can be applied in real-world business scenarios. Additionally, the use of AI-assisted tools enhanced efficiency and reduced manual effort.

From this fellowship experience, I gained strong practical knowledge in data analysis, data visualization, and the use of modern tools like Excel and Power BI. Overall, this project helped in developing both technical and analytical skills required for future career growth.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

APPENDICES

Appendix A — Sample Dataset (Raw Dataset Summary):

This appendix presents a sample of the raw dataset used for analysis. The dataset contains 1000 rows and includes key variables such as Date, Region, City, Product Name, Customer Type, Quantity, Unit Price, Total Sales, Total Cost, Net Profit, Discount Percentage, Payment Mode, Marketing Campaign, Customer Rating, and Validated Profit.

Below is a preview of selected rows from the dataset:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Date	Region	City	Product_Name	Customer_Type	Quantity	Unit_Price	Total_Sales	Total_Cost	Net_Profit	Discount_Percentage	Payment_Mode	Marketing_Campaign	Customer_Rating	Validated_Profit
2	2025-08-21	West	Chennai	Medicines	Wholesale	11	17744	195184	153457	41727	10	Cash	No Campaign	4.1	41727
3	2025-12-16	East	Hyderaba	Laptop	Wholesale	2	16288	32576	22269	10307	2	Net Banking	Festival Offer	3.3	10307
4	2025-09-08	West	Hyderaba	Rice Bags	Wholesale	10	30371	303710	236862	66848	21	Card	Summer Sale	3.1	66848
5	2025-11-16	East	Mumbai	Sofa	Wholesale	14	43789	613046	470923	142123	21	Card	Summer Sale	3.6	142123
6	2025-02-05	East	Delhi	Refrigerator	Corporate	1	31551	31551	20167	11384	13	Card	Clearance Sale	4.3	11384
7	2025-12-02	West	Hyderaba	Medicines	Corporate	7	5497	38479	24460	14019	16	Card	No Campaign	3.8	14019
8	2025-08-28	North	Hyderaba	Mobile	Retail	11	49305	542355	357221	185134	6	UPI	Clearance Sale	5	185134
9	2025-01-02	North	Hyderaba	Medicines	Corporate	12	38025	456300	350364	105936	15	Cash	Festival Offer	4.1	105936
10	2025-02-07	North	Chennai	Rice Bags	Corporate	5	32319	161595	102229	59366	5	Net Banking	No Campaign	4.8	59366
11	2025-05-11	West	Bangalore	T-Shirts	Retail	8	38645	309160	213304	95856	15	Net Banking	Festival Offer	4.1	95856
12	2025-01-04	West	Hyderaba	Mobile	Retail	20	1645	32900	21555	11345	23	UPI	New Launch	4.9	11345
13	2025-10-23	East	Delhi	Refrigerator	Retail	4	42465	169860	137377	32483	16	Net Banking	No Campaign	3.9	32483
14	2025-03-10	North	Bangalore	Refrigerator	Corporate	8	49014	392112	253753	138359	23	UPI	No Campaign	4.2	138359
15	2025-03-06	South	Kolkata	T-Shirts	Corporate	11	42418	466598	281870	184728	13	Card	No Campaign	4.2	184728
16	2025-03-17	South	Mumbai	Sofa	Wholesale	12	8021	96252	84126	12126	9	Net Banking	New Launch	4.8	12126
17	2025-11-24	South	Delhi	Laptop	Corporate	5	17695	88475	64770	23705	9	UPI	New Launch	3.3	23705
18	2025-11-07	North	Chennai	Refrigerator	Corporate	7	37636	263452	186162	77290	22	UPI	Clearance Sale	3.9	77290
19	2025-02-13	West	Delhi	Rice Bags	Wholesale	2	7633	15266	13623	1643	22	Cash	Clearance Sale	4.9	1643
20	2025-10-22	South	Bangalore	Mobile	Retail	3	13522	40566	28963	11603	21	Cash	Clearance Sale	4.5	11603
21	2025-09-25	West	Delhi	T-Shirts	Wholesale	14	5671	79394	51852	27542	13	UPI	Clearance Sale	4.1	27542
22	2025-02-26	South	Bangalore	Laptop	Wholesale	10	40794	407940	350406	57534	3	Card	No Campaign	3.5	57534
23	2025-06-14	West	Mumbai	Medicines	Wholesale	5	7723	38615	26014	12601	6	Card	Clearance Sale	3.7	12601
24	2025-05-11	East	Kolkata	Mobile	Corporate	14	48336	676704	490473	186231	15	Card	No Campaign	3	186231
25	2025-07-22	West	Chennai	T-Shirts	Retail	12	19203	230436	169331	61105	10	Cash	Summer Sale	4.6	61105
26	2025-04-08	East	Kolkata	Refrigerator	Corporate	6	33365	200190	136523	63667	1	UPI	New Launch	3.5	63667

Note: The complete dataset is available in the project files.

Appendix B-Cleaned Data Summary:

- The dataset was cleaned by removing missing values and correcting inconsistent entries.
- Duplicate records were identified and removed to ensure data accuracy.
- Column names were standardized and formatted properly.
- Data types (such as date and numeric fields) were corrected for analysis.
- Outliers and irrelevant data were handled to improve reliability.
- The final cleaned dataset is structured, consistent, and ready for visualization and analysis.

Appendix C — Key Excel Formulas / DAX Measures Used:

The formulas, functions and DAX measures used in the project are;

- VLOOKUP / XLOOKUP: Used to retrieve product or region-related information from datasets.
- IFERROR: Used to handle missing or invalid data values during cleaning.
- COUNTIF: Used to identify and count duplicate records in the dataset.
- SUM / AVERAGE: Used for calculating total sales, cost, and average rating.

DAX Measures (Power BI):

- Total Sales = SUM (Sales [Total_Sales])
- Total Cost = SUM (Sales [Total_Cost])
- Total Profit = SUM (Sales [Net_Profit])
- Profit Margin % = DIVIDE ([Total Profit], [Total Sales]) * 100
- Average Rating = AVERAGE (Sales [Customer_Rating])
- Discount Range =

```
SWITCH (
TRUE (),
Sheet1[Discount_Percentage] <= 10, "0-10",
Sheet1[Discount_Percentage] <= 20, "10-20",
Sheet1[Discount_Percentage] <= 30, "20-30",
Sheet1[Discount_Percentage] <= 40, "30-40",
"40+")
```

Appendix D— AI Prompts Used:

Prompt 1: “Clean and structure this sales dataset for better analysis and remove inconsistencies.”

Prompt 2: “Suggest suitable charts and dashboards for analyzing sales, profit, and customer trends in Power BI.”

Prompt 3: “Generate key insights and findings from sales data including trends, top products, and regional performance.”

Prompt 4: “Help in writing report sections like conclusion, findings, and advantages for my data analytics project.”